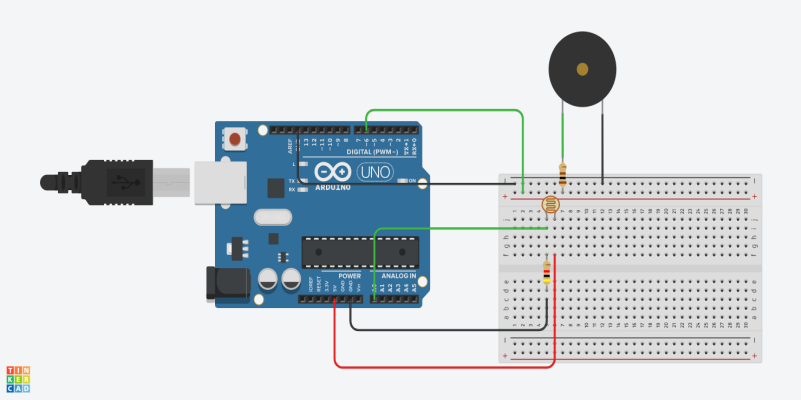
CSD 404 Internet of Things Lab 3

Name: Prakhar Rathi
Roll Number: 1810110169

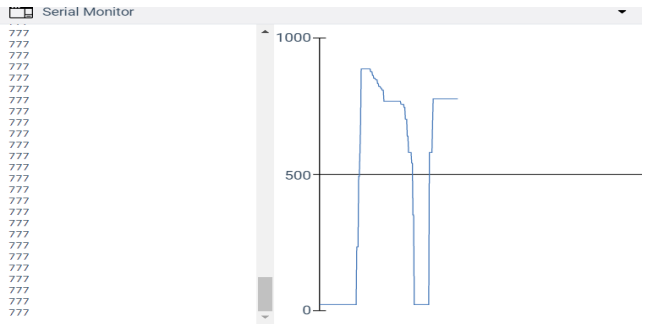
For this lab assignment, you will write basic Arduino programs.

Questions - Put the circuit diagram, code, and serial monitor output (Q3) after each question

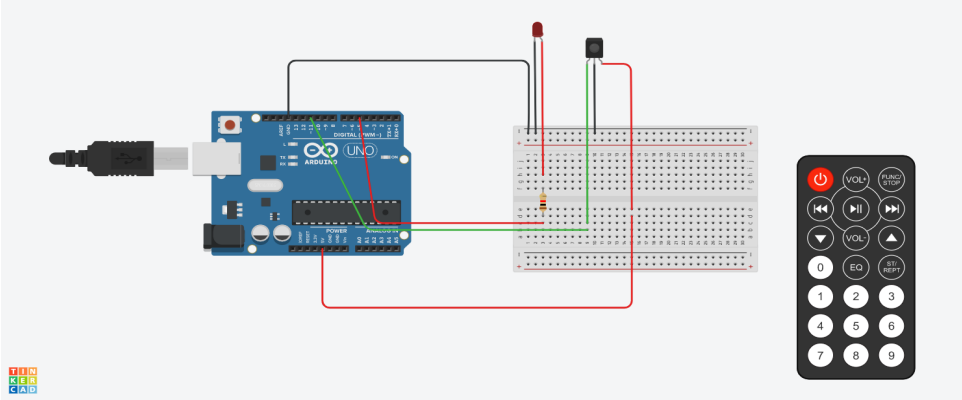
1. Write a sketch using LDR, transistor, and Piezo Buzzer to measure the intensity of the light. If its analog value crosses 600, generate a buzzer alarm for 2 seconds. Send the LDR values to the Serial Monitor also.




```
int sensorReading = 0;
int flag = 0;

void setup()
{
  // Set up pins
  pinMode(A0, INPUT);
  Serial.begin(9600);
  pinMode(6, OUTPUT);
}

void loop()
{
  sensorReading = analogRead(A0);
  Serial.println(sensorReading);

  if(sensorReading > 600 && flag == 0){
    tone(6, 100);

    // Play for 2 seconds
    delay(2000);
    noTone(6);
    delay(20);
    flag = 1;
  }
  if(sensorReading < 600)
    flag = 0;
}
```



2. Write a sketch using an LED, IR sensor, and IR Remote. On pressing the button “1” LED should light up, and on pressing the button “2”, LED should become off.



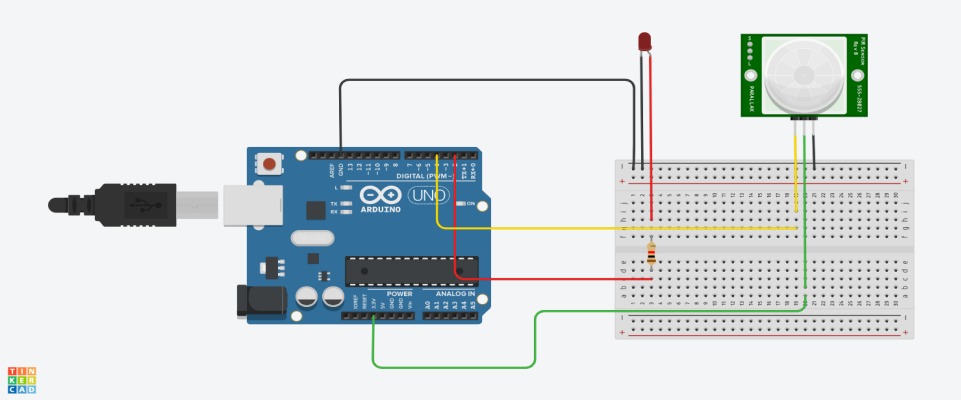
Simulation Code below

```
#include <IRremote.h>

//Define Pin
int led = 5;
int REC_PIN = 11;

//IR Library stuff
IRrecv irrecv(REC_PIN);
decode_results results;


void setup()
{
  //Set Led Pin
  pinMode(led, OUTPUT);

  // Enable serial monitor
  Serial.begin(9600);
  irrecv.enableIRIn();
}

void loop() {
  if (irrecv.decode(&results)) {
    unsigned int value = results.value;
    Serial.println(value);

    switch (value) {
      case 2295:
        digitalWrite(led, HIGH);
        delay(5);
        break;

      case 34935:
        digitalWrite(led, LOW);
        delay(5);
        break;

    }

    irrecv.resume();
  }
}
```

Serial Monitor

```
34935
2295
2295
34935
2295
34935
2295
34935
```

3. Write an Arduino sketch using a PIR Sensor, and an LED. In case of a moving object detection the LED should light up. If object become out of the range of the sensor, the LED should become off. Send a message to Serial Monitor in both the cases.




```
int sensorState = 0;

void setup()
{
  // Set up pins
  pinMode(4, INPUT);
  pinMode(2, OUTPUT);
  Serial.begin(9600);
}

void loop()
{

  // read the state of the sensor
  sensorState = digitalRead(4);

  // check if sensor pin is HIGH. if it is, set the LED on.
  if (sensorState == HIGH) {
    digitalWrite(2, HIGH);
    Serial.println("Sensor activated");
    delay(500);

  } else {
    digitalWrite(2, LOW);
    Serial.println("Sensor deactivated");
    delay(500);
  }

  delay(10);

}
```

Serial Monitor

```
Sensor activated
Sensor activated
Sensor activated
Sensor aSensor deactivated
Sensor deactivated
Sensor activated
Sensor activated
Sensor deactivated
Sensor activated
Sensor activated
Sensor deactivated
Sensor activated
Sensor activated
Sensor deactivated
Sensor deactivated
Sensor deactivated
Sensor deactivated
Sensor deactivated
Sensor activated
Sensor activated
Sensor deactivated
```